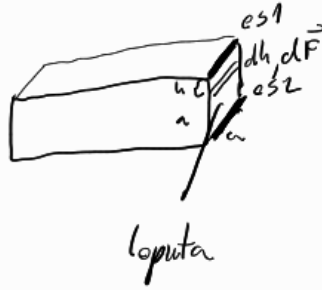


Hidrostatika

29/1

$$a = 10 \text{ cm}$$

$$M_{1,2} = ?$$



$$p = \rho g h = \frac{F}{S}$$

$$F = \rho g h S$$

$$dF = \rho g h \cdot dS = \rho g h \cdot a \cdot dh$$

$$dM_2 = dF(a-h) = a \rho g h(a-h) dh$$

$$M_2 = a \rho g \int_0^a (ha - h^2) dh =$$

$$= a \rho g \left(\frac{a^3}{2} - \frac{a^3}{3} \right) = \underline{\underline{\frac{1}{6} \rho g a^4}}$$

$$dM_1 = dF \cdot h = a \rho g h^2 dh$$

$$M_1 = \int_0^a a \rho g h^2 dh = a \rho g \frac{a^3}{3} = \underline{\underline{\frac{1}{3} \rho g a^4}}$$

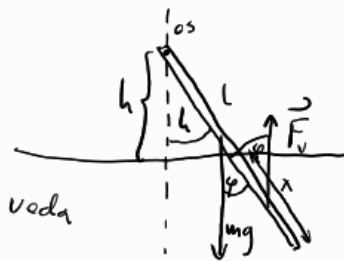
29/10

$$l = 1 \text{ m}$$

$$\rho = 700 \frac{\text{kg}}{\text{m}^3}$$

$$h = 0,4 \text{ m}$$

$$\varphi = ?$$



$$M_v = F_v \left(l - \frac{x}{2} \right) \sin \varphi$$

$$M_g = mg \frac{l}{2} \sin \varphi$$

$$F_v = \rho_v g V_x = \rho_v g S x$$

$$h = \rho \cdot S \cdot l$$

$$M_v = M_g$$

$$\rho_v x \left(l - \frac{x}{2} \right) \sin \varphi = \rho l \frac{l}{2} \sin \varphi$$

$$2x \left(l - \frac{x}{2} \right) = \frac{\rho}{\rho_v} \frac{l^2}{2}$$

$$x^2 - 2lx + \frac{\rho}{\rho_v} l^2 = 0$$

$$x_{1,2} = \frac{2l \pm \sqrt{4l^2 - 4 \frac{\rho}{\rho_v} l^2}}{2} = l \left(1 \pm \sqrt{1 - \frac{\rho}{\rho_v}} \right)$$

$$\rightarrow x = l \left(1 - \sqrt{1 - \frac{\rho}{\rho_v}} \right) = 0,45 \text{ m}$$

$$\cos \varphi = \frac{d}{l-x} \rightarrow \underline{\underline{\varphi = 43^\circ}}$$

29/6
 $m_p = 100g$

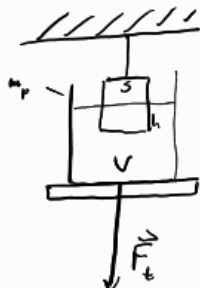
$V = 1l$

$h = 5cm$

$S = 60cm^2$

$\rho_{rel} = 2,7 g/cm^3$

$F_t = ?$



$$\begin{aligned} F_t &= m_p g + m_v g + F_{vzcg} = \\ &= m_p g + V \cdot \rho_v \cdot g + S h \rho_v g = \\ &= g(m_p + V \rho_v + S h \rho_v) \approx 14N \end{aligned}$$

29/14

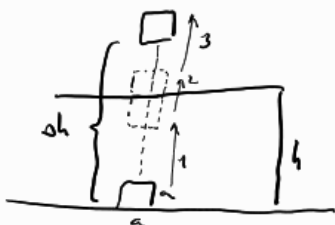
$h = 6m$

$a = 1m$

$\rho_a = 2,5 g/cm^3$

$oh = 10m$

$A = ?$



$$A = \int \vec{F} \cdot d\vec{s}$$

① $A_1 = F_1 \cdot s_1 = (F_g - F_v)(h-a) = g(a^3 \rho_a - a^3 \rho_v)(h-a) =$
 $= \underline{a^3 g (\rho_a - \rho_v)(h-a)}$

③ $A_3 = F_3 \cdot s_3 = F_g(oh-h) = \underline{\rho_a a^3 g(oh-h)}$

②



$z_2 = a$

$$\begin{aligned} F_2 &= F_g - F_{vzg} = g \rho_a a^3 - (a^2 x g \rho_v) = \\ &= g a^2 (\rho_a a - x \rho_v) \end{aligned}$$

$$A_2 = - \int_a^0 F_2 \cdot dx = - g a^2 \int_a^0 (\rho_a a - x \rho_v) dx =$$

$$= g a^2 \left(\rho_a a x - \frac{\rho_v}{2} x^2 \right) \Big|_a^0 =$$

$$= g a^2 \left(a^2 \frac{\rho_v}{2} - a^2 \rho_a \right) =$$

$$= \underline{g a^4 (\rho_a - \rho_v/2)}$$

$$\begin{aligned} A &= A_1 + A_2 + A_3 = a^3 g (\rho_a - \rho_v)(h-a) + \rho_a (oh-h) + a (\rho_a - \rho_v/2) = \\ &= a^3 g (\rho_a h - \rho_a a - \rho_v h + \rho_v a + \rho_a oh - \rho_a h + \rho_a a - \frac{a}{2} \rho_v) = \\ &= \underline{a^3 g (\rho_a oh - \rho_v (h - \frac{a}{2}))} \end{aligned}$$

A , če bi drigornili
 po znaku

Delo, če voda
 zleze dol iz glatine
 na mesto, kjer je bila kocka na račitku

Hydrodynamika

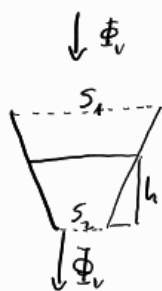
8.2.2

$$S_1 = 3 \text{ cm}^2$$

$$S_2 = 1 \text{ cm}^2$$

$$\Phi_v = 0,1 \frac{\text{l}}{\text{s}}$$

$$h = ?$$



$$\Phi_v = v_1 S_1 = v_2 S_2$$

$$v_1 = \frac{0,1 \frac{\text{dm}^3}{\text{s}}}{3 \cdot 10^{-2} \text{ dm}^2} = 3,3 \frac{\text{dm}}{\text{s}} = 0,33 \frac{\text{m}}{\text{s}}$$

$$v_2 = \frac{0,1 \frac{\text{dm}^3}{\text{s}}}{1 \cdot 10^{-2} \text{ dm}^2} = 10 \frac{\text{dm}}{\text{s}} = 1 \frac{\text{m}}{\text{s}}$$

po enak pri obeh

$$\Rightarrow \rho g h_1 + \frac{\rho v_1^2}{2} = \rho g h_2 + \frac{\rho v_2^2}{2}$$

$$h = \frac{1}{2g} (v_2^2 - v_1^2) = \frac{1}{2g} \left(\left(\frac{\Phi_v}{S_2} \right)^2 - \left(\frac{\Phi_v}{S_1} \right)^2 \right) = \frac{\Phi_v^2}{2g} \left(\frac{1}{S_2^2} - \frac{1}{S_1^2} \right) = 1,4 \text{ cm}$$

8.2.3

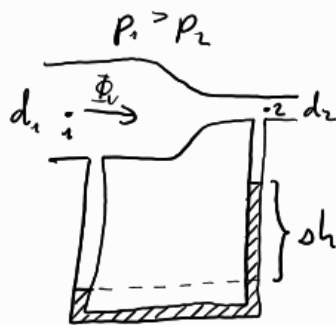
29/5

$$d_1 = 2 \text{ cm}$$

$$d_2 = 1,8 \text{ cm}$$

$$\rho_s = 1,02 \text{ g/cm}^3$$

$$\Delta h = 25 \text{ cm}$$



$$\Phi_v = S_1 v_1 = S_2 v_2$$

$$p_1 + \frac{\rho_s v_1^2}{2} = p_2 + \frac{\rho_s v_2^2}{2}$$

$\Phi_v = ?$

$$p_1 - p_2 = \Delta p = (\rho_s - \rho_v) g \Delta h$$

$$(\rho_s - \rho_v) g \Delta h = \frac{\rho_v}{2} (v_2^2 - v_1^2) = \frac{\rho_v \Phi_v^2}{2} \left(\frac{1}{S_2^2} - \frac{1}{S_1^2} \right)$$

$$\Phi_v = \sqrt{\frac{g \Delta h 2 (\rho_s - \rho_v)}{\rho_v \left(\frac{1}{S_2^2} - \frac{1}{S_1^2} \right)}} = 0,135 \frac{\text{l}}{\text{s}}$$

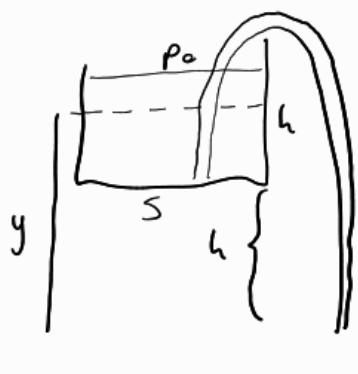
8.2.5

$$S = 100 \text{ cm}^2$$

$$h = 10 \text{ cm}$$

$$S' = 1 \text{ cm}^2$$

$$t = ?$$



$$\rho g h_1 + \frac{\rho v_1^2}{2} = \rho g h_2 + \frac{\rho v_2^2}{2}$$

$$v_2^2 = 2gy + v_1^2$$

$$S v_1 = S' v_2$$

$$v_1^2 = 2gy + \left(\frac{S}{S'} \right)^2 v_1^2$$

$$v_2 = \frac{S}{S'} v_1$$

$$v_1^2 \left(\left(\frac{S}{S'} \right)^2 - 1 \right) = 2gy$$

$$v_1 = \sqrt{\frac{2gy}{\left(\frac{S}{S'} \right)^2 - 1}}$$

$$-\frac{dy}{dt} = \sqrt{\frac{2g}{\left(\frac{S}{S'} \right)^2 - 1}} \sqrt{y}$$

$$\int_h^{2h} y^{-\frac{1}{2}} dy = \int_0^t \sqrt{\frac{2g}{(\frac{y}{h})^2 - 1}} dt$$

$$2\sqrt{y} \Big|_h^{2h} = t \sqrt{\frac{2g}{(\frac{y}{h})^2 - 1}}$$

Za max višino ceni zgoraj!

$$p_3 + \rho g h_{\max} + \frac{\rho v_3^2}{2} = p_0 + \frac{\rho v_0^2}{2}$$

$$2\sqrt{h}(2-1) = t \sqrt{\frac{2g}{(\frac{y}{h})^2 - 1}}$$

$$t = \frac{2\sqrt{g h} (2-1)}{\sqrt{\frac{2g}{5^2 - 1}}} = 5,9s$$

$$\rho g h_{\max} = p_0 - p_3$$

$$h_{\max} = \frac{p_0}{\rho g} = 10m$$

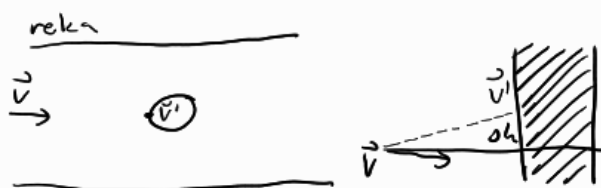
o, v realnosti uporabimo izparilni tlak vode
ZA VODO

8.2.4

$$v' = 0$$

$$\Delta h = 20cm$$

$$v = ?$$



$$\frac{\rho v^2}{2} = \rho g \Delta h$$

$$v = \sqrt{2g \Delta h} = \sqrt{4 \frac{m^2}{s^2}} = 2 \frac{m}{s}$$

8.3.1 $h = 10m$

$$\eta = 9,001 kg/s$$

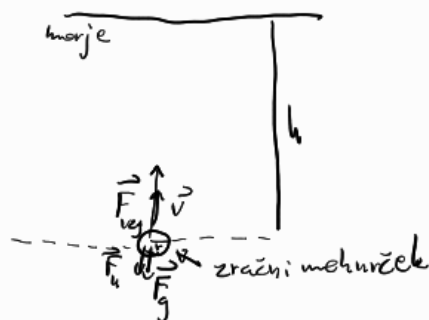
$$\rho_w = 2,4 \frac{kg}{m^3}$$

$$C_u = 0,1$$

$$v(r) = ?$$

$$r_{maj} = ?$$

kje je kvadratni zakon ok



$$F_{vez} = (F_g + F_b)$$

$$\frac{4\pi r^3}{3} \rho_{H_2O} g = \frac{4\pi r^3}{3} \rho_w g + \frac{1}{2} C_u \rho_{H_2O} v^2 S$$

$$g \cdot \frac{4\pi}{3} r^3 (\rho_{H_2O} - \rho_w) = \frac{1}{2} C_u \rho_{H_2O} v^2 \pi r^2$$

$$v = \sqrt{\frac{8g(\rho_{H_2O} - \rho_w)r}{3C_u \rho_{H_2O}}}$$

predp, da smo na $h=10m$ že v ravnotežju

$$Re = \frac{\rho v d}{\eta} = \begin{cases} > 10^3 \rightarrow \text{kvadratni} \\ < 10^2 \rightarrow \text{linearni} \end{cases}$$

$$\frac{\rho_{H_2O} \cdot 2r^{\frac{3}{2}} \cdot 2}{\eta} \cdot \sqrt{\frac{2g(\rho_{H_2O} - \rho_w)r}{3C_u \rho_{H_2O}}} > 10^3$$

$$r^{\frac{3}{2}} > \frac{10^3 \cdot \eta}{\rho_{H_2O} \cdot 4} \sqrt{\frac{3C_u \rho_{H_2O}}{2g(\rho_{H_2O} - \rho_w)}}$$

$$r > 1,5mm$$

Bolj kot je površina gladka, večje

Re rabimo za kvadratni zakon upora

29/17

$$m = 200 \text{ kg}$$

$$S = 2 \text{ m}^2$$

$$v_v = 40 \text{ km/h}$$

$$v_k = 10 \text{ km/h}$$

$$\rho_{\text{az}} = 1,2 \frac{\text{kg}}{\text{m}^3}$$

$$C_u = 0,8$$

$$t_k = ?$$



$$F_u = ma$$

$$\frac{1}{2} C_u \rho S (v_v - v)^2 = m \frac{dv}{dt}$$

$$v_v - v = u$$

$$-dv = du$$

$$\int_0^{t_k} k dt = \int_{v_k}^{v_v} \frac{dv}{(v_v - v)^2}$$

$$k \cdot t_k = \int_0^{v_k} \frac{-du}{u^2} = \frac{1}{v_v - v} \Big|_0^{v_k} = \frac{1}{v_v - v_k} - \frac{1}{v_v}$$

$$t_k = \left(\frac{v_v - v_k}{(v_v - v_k) v_v} \right) \frac{2m}{C_u \rho S} = 6,25 \text{ s}$$

8.3.3

$$v_v = 10 \text{ m/s}$$

$$S_1 = S_2$$

$$P_1 = P_2$$

$$t_1 = 2t_2$$

$$v_1 = ?$$

(1)



(2)



$$2v_1 = v_2$$

$$P_1 = P_2$$

$$(v_v + v_1)^2 \cdot \frac{1}{2} = v_2^2 \cdot \frac{1}{2} = 8v_1^2$$

$$7v_1^2 - 2v_v \cdot v_1 - v_v^2 = 0$$

$$v_1 = 5,5 \text{ m/s}$$

8.4.1

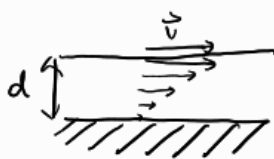
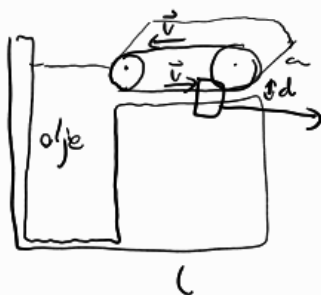
$$d = 1 \text{ cm}$$

$$a = 2 \text{ m}$$

$$l = 10 \text{ m}$$

$$\eta = 0,6 \frac{\text{kg}}{\text{ms}}$$

$$v = 0,5 \text{ m/s}$$



$$\Phi_v = a \cdot d \cdot \frac{1}{2} v = 0,01 \cdot \frac{1}{2} \cdot \frac{1}{2} \frac{\text{m}^3}{\text{s}} = 0,005 \frac{\text{m}^3}{\text{s}}$$

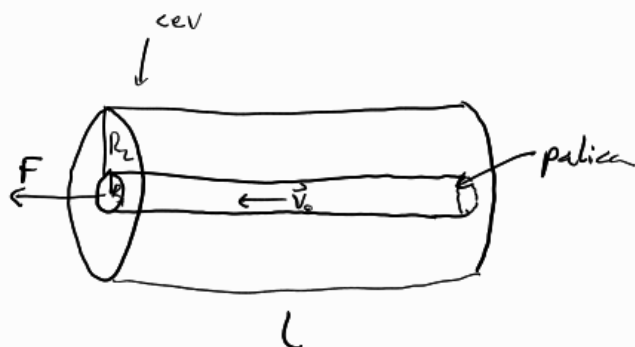
$$P = F_u \cdot v = \eta S' \frac{dv}{dy} \cdot v = \eta l \cdot a \cdot \frac{v}{d} \cdot v = \frac{\eta v^2 l a}{d} = 300 \text{ W}$$

$$\Phi_v = ?$$

$$P = ?$$

8.4.3

R_2
 l
 η
 $R_1 \ll R_2$
 $\vec{v}_0$
 $F = ?$
 $v(r) = ?$

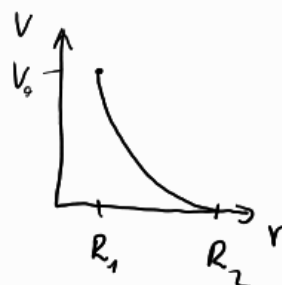


$$F_u = \eta S \frac{dv}{dr} = -\eta 2\pi r l \frac{dv}{dr}$$

$$\int_{R_1}^r -\frac{F_u}{\eta 2\pi l} \cdot \frac{1}{r} dr = \int_{v_0}^v dv$$

$$\frac{F_u}{\eta 2\pi l} \cdot \ln \frac{r}{R_1} = v_0 - v$$

$$v(r) = v_0 - \frac{F_u}{2\pi \eta l} \ln \frac{r}{R_1}$$



$$r = R_2: v(R_2) = 0$$

$$v_0 = \frac{F_u}{2\pi \eta l} \ln \frac{R_2}{R_1} \rightarrow$$

$$F = \frac{2\pi \eta l \cdot v_0}{\ln \frac{R_2}{R_1}}$$

Poiseuille zakon (za tok tekočine po cevi)

$$① v(r) = \frac{\Delta p}{4\eta l} (R^2 - r^2)$$

$$② v_{\max} = \frac{\Delta p R^2}{4\eta l}$$

$$③ \Phi_v = \frac{\Delta p \pi R^4}{8\eta l}$$

8.4.4)

$$h = 20 \text{ cm}$$

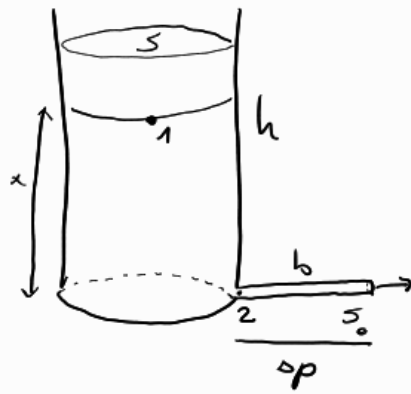
$$S = 20 \text{ cm}^2$$

$$b = 10 \text{ cm}$$

$$S_0 = 0,2 \text{ mm}^2$$

$$\eta = 8,9 \cdot 10^{-4} \frac{\text{kg}}{\text{ms}}$$

$$t_{1/2} = ?$$



$$\Phi_v = \text{konst.}$$

$$S v_1 = S_0 v_2$$

$$v_2 = \frac{S}{S_0} v_1 \rightarrow v_2 \gg v_1$$

Bernoulli zu 1-2:

$$\cancel{p} + \rho g x + \frac{\rho v_1^2}{2} = \cancel{p_0} + \Delta p + \frac{\rho v_2^2}{2} \rightarrow 0$$

$$\frac{\rho v_2^2}{2} \approx \frac{10^3 \cdot 1}{2 \pi^2} \approx 50 \frac{\text{N}}{\text{m}^2}$$

$$\rho g h = 10^3 \cdot 10 \cdot 0,2 = 2000 \frac{\text{N}}{\text{m}^2}$$



$$\Delta p = \rho g x$$

$$\text{event: } \Delta p_{\text{max}} = \rho g h$$

$$v_{2\text{max}} = \frac{\Delta p R_0^2}{4 \eta b} = \frac{\rho g h R_0^2 \pi S_0}{4 \eta b \pi}$$

$$= \frac{10^3 \cdot 10 \cdot 0,2 \cdot 0,2 \cdot 10^{-6} \cdot 10^{-2}}{4 \cdot 10^{-4} \cdot 0,1 \cdot \pi} = \frac{1}{\pi} \frac{\text{m}}{\text{s}}$$

$$\Phi_v = \cancel{p_0} v_2 = \frac{\Delta p \pi R_0^4 \pi S_0^2}{8 \eta b \pi} \cdot \rho g x$$

$$\frac{dx}{dt} = \frac{\rho g x S_0^2}{8 \pi \eta b S}$$

$$\int_h^{\frac{h}{2}} -\frac{1}{x} dx = \frac{\rho g S_0^2}{8 \pi \eta b S} \int_0^{t_{1/2}} dt$$

$$\frac{8 \pi \eta b S}{\rho g S_0^2} \cdot \ln \frac{h}{h/2} = \boxed{t_{1/2} = \frac{8 \pi \eta b S}{\rho g S_0^2} \ln 2} \approx 14 \text{ s}$$

29/11

$$2r = 1 \text{ cm}$$

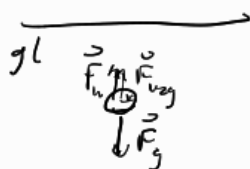
$$\rho_{st} = 2,5 \cdot 10^3 \text{ kg/m}^3$$

$$\rho_{gl} = 126 \text{ g/cm}^3$$

$$\eta_{gl} = 1,39 \text{ kg/s}$$

$$S(0,99 v_k) = ?$$

$$V_k = \frac{g \frac{4\pi r^2}{3} (\rho_{st} - \rho_{gl})}{6\eta_{gl} / 2g} = \frac{2r^2 (\rho_{st} - \rho_{gl}) g}{9\eta_{gl}}$$



$$Q_e = \frac{\rho_{gl} v d}{\eta}$$

$$F_u = 6\pi \eta r v$$

$$F_u + F_{vz} - F_g = -m \frac{dv}{dt}$$

$$6\pi \eta r v + \rho_{gl} g \frac{4\pi r^3}{3} - g \cdot \rho_{st} \frac{4\pi r^3}{3} = -m \frac{dv}{dt} \cdot \frac{dx}{dx} v$$

$$\leftarrow g \frac{4\pi r^3}{3} (\rho_{st} - \rho_{gl}) - 6\pi \eta r v = m v \frac{dv}{dx} \cdot \frac{4\pi r^3}{3}$$

$$g \cdot 2r^2 (\rho_{st} - \rho_{gl}) - 9\eta v = v \rho_{st} 2r^2 \frac{dv}{dx}$$

$$\int_0^S dx = \int_0^{0,99 v_k} \frac{v \cdot \rho_{st} 2r^2}{2gr^2 (\rho_{st} - \rho_{gl}) - 9\eta v} dv =$$

$$\frac{2gr^2 (\rho_{st} - \rho_{gl}) - 9\eta v}{9\eta} = g_y (v_k - v)$$

$$S = \int_0^{0,99 v_k} \frac{v \cdot \rho_{st} 2r^2}{9\eta (v_k - v)} dv = \frac{\rho_{st} 2r^2}{9\eta} \int_0^{0,99 v_k} \frac{v}{v_k - v} dv =$$

$$v_k - v = u$$

$$-dv = du$$

$$= \frac{\rho_{st} 2r^2}{9\eta} \int_0^{0,99 v_k} \frac{u - v_k}{u} du =$$

$$= \frac{\rho_{st} 2r^2}{9\eta} \left(0,99 v_k - v_k \ln(1 - 0,99) \right) =$$

$$= \frac{\rho_{st} 2r^2 v_k}{9\eta} \left(0,99 - \underbrace{\ln(0,01)}_{-\ln 100} \right) = 1,8 \text{ mm}$$

Površinska napetost

$$\gamma = \dots$$

$$F_{\text{pov}} = \gamma l$$

$$W_{\text{pov}} = \gamma S$$

8.5.1

(napisano mehurček, koliko dela opravi)

MEHURČEK IMA 2 POUČINI

r_1
 r_2
 $A = ?$



1. PRER. ENERGIJ

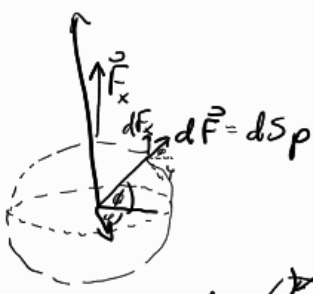
$$A = W_{\text{por2}} - W_{\text{por1}} = \gamma (S_2 - S_1) = \gamma (4\pi(r_2^2 - r_1^2)) = 8\pi\gamma(r_2^2 - r_1^2)$$

2. PRER. SIL

$$A = \int F ds = - \int p dV$$

$$V = \frac{4\pi r^3}{3}$$

$$dV = 4\pi r^2 dr$$



$r \cos \theta d\varphi$ $ds = r^2 \cos \theta d\theta d\varphi$
 $d\varphi$ $r d\theta$ $dF = r^2 \cos \theta d\theta d\varphi \cdot p$

$$dF_y = dF \cdot \sin \theta$$

$$F_x = pr^2 \int_0^{\pi/2} \cos \theta \sin \theta d\theta \int_0^{2\pi} d\varphi =$$

$$= pr^2 \int_0^{\pi/2} \frac{1}{2} \sin(2\theta) d\theta = \pi pr^2 \left(\frac{1}{2} - (-\frac{1}{2}) \right) = \underline{\underline{\pi pr^2}} = \gamma \cdot 2\pi r \cdot 2$$

$$p = \frac{\gamma \cdot 4}{r}$$

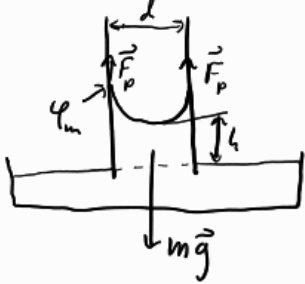
$$A = \int p dV = -4\gamma \int_{r_1}^{r_2} \frac{4\pi r^2 dr}{r} = -16\pi\gamma \left(\frac{r_2^2}{2} - \frac{r_1^2}{2} \right) = -8\pi\gamma(r_2^2 - r_1^2)$$

je neg., ker je to delo, ki ga opravi plin v mehurčku

8.5.2 dvig vode med steklenima ploščama

za čisto steklo:

$d = 1 \text{ mm}$
 $\gamma = 0,07 \frac{\text{N}}{\text{m}}$
 $\varphi_m = 0$
 $h = ?$



$$mg = \rho \cdot h \cdot d \cdot l \cdot g$$

$$F_p = l \gamma \cdot \cos \varphi_m$$

$$2\gamma \cos \varphi_m = \rho h d g$$

$$h = \frac{2\gamma}{\rho d g} = \frac{2 \cdot 0,07 \text{ N}}{1000 \frac{\text{kg}}{\text{m}^3} \cdot 10^{-3} \text{ m} \cdot 10 \frac{\text{m}}{\text{s}^2}} = 14 \text{ mm}$$

→ za okroglo cevko

$$\rho \cdot h \cdot \pi r^2 g = 2\pi r \gamma \cos \varphi_m$$

8.5.3

voda na parafinu

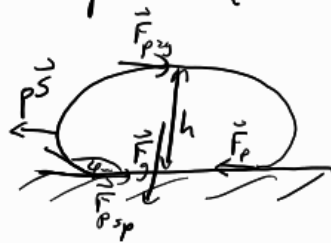
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$$\gamma = 0,07 \frac{\text{N}}{\text{m}}$$

$$\varphi_m = 180^\circ$$

velika kaplja

$$h = ?$$



$$p = \frac{F}{S} = \rho g h$$

$$F_{sy} - F_{sp} \cos \varphi_m = pS$$

$$2\gamma l = \rho g \frac{h^2}{2} l$$

obseg

$$h = \sqrt{\frac{4\gamma}{\rho g}} = \sqrt{\frac{4 \cdot 0,07 \frac{\text{N}}{\text{m}}}{1000 \frac{\text{kg}}{\text{m}^3} \cdot 10 \frac{\text{m}}{\text{s}^2}}} = \frac{2\sqrt{0,07}}{10^2} \text{ m}$$

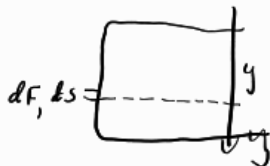
$$\approx 5 \mu\text{m}$$

$$p =$$

$$dF = p dS =$$

$$= \rho g y l dy$$

$$F = \rho g \frac{h^2}{2} l$$



z energijami:

$$W = W_{\text{pot}} + W_{\text{pot}} = m g \frac{h}{2} + S' \gamma_{pk} + S' \gamma_{kt}$$

$$W_{\text{pot}} = S \gamma$$

$$V = S' h = \frac{m}{\rho}$$

$$\rightarrow S' = \frac{V}{h} = \frac{m}{\rho h}$$

$$W = m g \frac{h}{2} + \frac{m}{\rho h} (\gamma_{pk} + \gamma_{kt})$$

$$\frac{\partial W}{\partial h} = \frac{mg}{2} - \frac{m}{\rho h^2} (\gamma_{pk} + \gamma_{kt}) = 0$$

$$h = \sqrt{\frac{2(\gamma_{pk} + \gamma_{kt})}{\rho g}}$$

